

Lab: Philosophy of Science Through Active Inference

Learning Goal: Apply Active Inference to analyze theory change, parsimony, and the nature of scientific progress.

Part 1: Theory Change Case Study

Exercise: Choose a historical theory change (e.g., geocentric → heliocentric, phlogiston → oxygen, Newtonian → Einsteinian). Analyze it in Active Inference terms:

Phase	Scientific Activity	Active Inference Analog
Normal science	Data collection within paradigm	Parameter learning (updating a)
Anomaly	Persistent unexplained observations	Rising prediction errors
Crisis	Paradigm inadequacy	High free energy, model evidence decline
Revolution	New paradigm	Bayesian model selection
Resolution	New normal science	Parameter learning in new model

Write a 300-word analysis of your chosen case.

Write your response here

Part 2: Occam's Razor in Action

Learning Goal: Apply the Complexity-Accuracy trade-off to real examples.

Exercise: Consider two competing models for the same data:

- Model A: 3 parameters, accuracy = 8.0
- Model B: 30 parameters, accuracy = 8.5
- Compute the approximate free energy for each ($F \approx \text{Complexity} - \text{Accuracy}$)
- Which model does Occam's Razor favor?
- Under what conditions would you prefer Model B?
- How does this relate to overfitting in machine learning?

Write your response here

Part 3: Demarcation Analysis

Learning Goal: Apply the Active Inference demarcation criterion to borderline cases.

Exercise: Classify each as scientific, pseudoscientific, or dogmatic using Active Inference criteria:

Claim	Classification	Model Properties
"The Earth orbits the Sun"		
"Astrology predicts personality"		
"Homeopathy works via water memory"		
"Evolution by natural selection"		
"This conspiracy explains everything"		

What distinguishes each category in terms of prior strength, prediction accuracy, and model evidence?

Write your response here

Part 4: Induction and Prior Justification

Exercise: The problem of induction asks: Why should we expect the future to resemble the past?

1. State the problem precisely
2. Explain Active Inference's pragmatic response
3. Is this response philosophically satisfying? What objections remain?
4. How does this connect to the problem of prior selection in Bayesian inference?

Write your response here

Part 5: Reflection

In 250 words, reflect: Does Active Inference provide a unified account of scientific method, or does it oversimplify the complexity of actual scientific practice?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Historical analysis	Theory change formalization
2	Quantitative reasoning	Occam's Razor
3	Case classification	Demarcation criterion
4	Philosophical argument	Problem of induction
5	Critical reflection	Limits of formalization